AKSHOBYA A S

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LinkedIn GitHub Portfolio

PROFILE

Information Science & Engineering undergraduate (2027 batch) who builds full-stack products end to end: MERN and Django applications with AI features, offline-first PWAs, and secure, tested backends. Strong foundations in Java, Python, C, Data Structures & Algorithms, OOP and DBMS.

EDUCATION

Bachelor of Engineering (B.E.) – Information Science & Engineering — **Expected 2027**
The National Institute of Engineering, Mysuru — Current CGPA: 7.60

Pre-University (PCMC) — **2023**
Creative PU College, Karkala — Percentage: 96.33%

TECHNICAL SKILLS

- **Languages:** Java, Python, C, JavaScript, SQL
- **Frontend:** React.js, Vite, Tailwind CSS, Zustand, HTML5, CSS3, PWA (Service Workers, IndexedDB)
- **Backend:** Node.js, Express.js, Django, REST APIs, JWT Authentication
- **Databases:** MongoDB, PostgreSQL, MySQL, SQLite, Redis
- **AI / ML:** OpenAI & Gemini APIs, RAG, TensorFlow/Keras (CNN), scikit-learn
- **Tools:** Git, GitHub, Docker, Docker Compose, Jest, Postman, Vercel

PROJECTS

Smart Farming Guidance System – B.E. Major Project (Team of 2) | *Django, React, TensorFlow, Gemini* Code

- Built a Django + React platform that diagnoses crop diseases from leaf photos using a **MobileNetV2 CNN** trained on 9 disease classes across black pepper, rice and arecanut, reaching **93.7% test accuracy**.
- Added a **Gemini-powered Kannada voice assistant**, a crop lifecycle tracker with weather-driven disease-risk alerts and a farming diary; every third-party API is routed through a Django proxy so no keys reach the client.

AI Semester Companion – AI Study Platform | *MERN, OpenAI, Redis, Docker* Code

- Built a **MERN** platform that turns an uploaded syllabus and notes into a unit-by-unit study roadmap, with AI lessons, adaptive quizzes, **SM-2 spaced-repetition** flashcards and doubt answers cited from the user's own files.
- Implemented **RAG** with local sentence-transformer embeddings and cosine search over MongoDB, **Redis** LLM caching, JWT auth with per-resource ownership checks, **28 Jest tests** and multi-stage **Docker** deployment.

Haajari – Offline-First School Attendance PWA | *MongoDB, Express, React, Node.js* Code

- Built a Kannada-first attendance app for Karnataka government schools with role-based access, spreadsheet import and one-tap marking that **works offline and syncs without duplicates** (IndexedDB + Workbox).
- Added parent absence alerts, QR pickup passes, live dashboards, mid-day meal counts and printable monthly registers (PDF/Excel); field-tested on a throttled low-end phone.

PitLog – Vehicle Fuel & Service Tracker | *Django, PostgreSQL, Chart.js, PWA* Code

- Built a Django PWA to log fuel, tank-to-tank mileage, service history and document expiry, with Chart.js analytics, CSV export, dark/light themes and offline support.
- Secured with email-OTP sign-up, login throttling with lockout and per-user data isolation, backed by **22 automated tests**.

ACHIEVEMENTS & ACTIVITIES

- Built **Namma Kannada** (Live) for the **Kannada Digital Innovation Hackathon 2026**: an interactive dialect map of Karnataka with 482 sourced words across 21 places and AI voice pronunciations.
- **Languages:** English, Kannada, Hindi